Prefix vacuity

The n_{stab}-mincut is prime-blind; `Iff.rfl` named the wrong object

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Abstract

Rounds 450–455 named the frequently-mincut as the n_{stab}-compression of $R^\dagger$ and proved that this prefix is exactly the archimedean fold through $J_P = \lceil \log 2/\Delta - 1 \rceil$. A worlds test with identical fold geometry (MAIN = von Mangoldt comb, ARCH = comb deleted) shows that every datum the prefix consumes is the same in both worlds to machine precision. The certified prefix positivity cannot tell a world with primes from one without, and therefore cannot imply RH. Lean never truncated the cone: `Iff.rfl` identified two Python charts by naming. The honest object is $R^\dagger$ at depth $\sim U_f/\Delta$, or the full cap for $\forall f$. No RH claim. No anti-RH claim.

Status. Ausgang `VACUOUS_CONFIRMED`. The r450–r455 prefix program is a reduction dictionary of the prime-blind zone (precedent r303), not mincut progress. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 Q1 — what Lean actually consumes

Proposition 1 (consumption map)**.** *The theorem `weil_nonneg_of_frequently_selected` (`FrequentlySelected.lean`:414) consumes, for each `GridElement` f:*

1. *one selected index k with $a_k \geq a_0(f)$ and $m_k \geq$ `meshExp` f (`selected_covers`, `Selected.lean`:412);*
2. *`selectedWindowConeSemidef` k (`FrequentlySelected.lean`:261–268,280): $R^\dagger(W_k^{\mathbb{R}}) \succeq \frac{1}{2} I$ at $n = \mathsf{cap} = (S{+}1)/2$ (`Selected.lean`:246), in a μ-ONB with $Q^T(\mathrm{combHankel})\,Q = I$ (`Selected.lean`:287) — the* full *window, including the comb;*
3. *the named bridge to plain `fullRead` ≥ 0 for every f with `meshExp` $f \leq m_k$ (`FrequentlySelected.lean`:322 325,358–362);*
4. *`fullRead` $=$ `archRead` $-$ `combRead` $+$ `poleRead` (`Elementwise.lean`:718–719), with*

$$\texttt{combRead}(a, f) = \sum_{n \in \mathrm{atoms}(a)} \mu_n\, f(\log n)$$

(`Elementwise.lean`:386–387).

The guard `meshExp` $f \leq m_k$ is a mesh-resolution *filter on the test function. It is not a grade cut of the window chain at n_{stab}. That is a category error.*

The r450 declaration `frequently_selected_prefix_augDualResolvent_ge_half` (`FrequentlySelected.l` 315) is definitionally the existing cone; `frequently_prefix_mincut_ident` is `Iff.rfl`. No truncation is stated, and none is proved. M_∂/ρ_0 never enters the Lean cone. It enters the Python plateau identity $q_* = M_\partial/(M_\partial + 5/7)$ (r452).

2 Q2 — the worlds test

Fix fold geometry (α, M, L, Δ). MAIN uses `arch_lags` + `prime_lags`. ARCH uses `arch_lags` only. Both keep the production `smooth_comb` border.

Lemma 1 (prefix is world-blind). *On kz69 ($n_{\mathrm{stab}} = 119 < J_P = 711$),*

$$\left|q^\dagger_{\mathrm{MAIN}}(119) - q^\dagger_{\mathrm{ARCH}}(119)\right| = 3.33 \cdot 10^{-16}.$$

On kz17 at depth 8 (strictly below $J_P = 19$), the same difference is $2.22 \cdot 10^{-16}$. Signed folded m_0 agrees to 10^{-14} (kz17: $m_0 = 2.5994580787932398$ in both worlds).

Lemma 2 (arithmetic onset is J_P). *MAIN leaves the plateau at the first prime tent. kz17: at $n = J_P = 19$, $\Delta q = 0.05166$ (pin); ARCH $q^\dagger$ stays at $q_* \approx 0.78566$. kz69: at $n = J_P = 711$, $\Delta q = 0.00401$; at the honest depth $n_f = \lceil U_f/\Delta \rceil = 2053$ for the tent $U_f = 2$, $\Delta q = 0.005698$.*

Lemma 3 (`combRead` sees primes). *For the even tent $F(u) = \max(0, 1 - |u|/2)$ (support past $\log 2$),*

$$\texttt{combRead} = \sum_{n \in \{2,3,4,5,7\}} \frac{2\Lambda(n)}{\sqrt{n}} F(\log n) = 1.7458508391553806$$

in MAIN, and 0 in ARCH. The same tent truncated at $U = 0.5 < \log 2$ gives 0 in both worlds. mpmath dps $= 50$ agrees with float64 to $2 \cdot 10^{-16}$.

The r455 "eventually per fixed n" argument is the opposite of coverage: as $\Delta \to 0$, every *fixed* grade falls into the blind zone $n < J_P(\Delta)$. It does not cover $n_f \sim U_f/\Delta$, which grows.

3 Q3 — the honest object

Proposition 2 (honest depth). *A test function of support U_f requires chain depth $n_f \sim U_f/\Delta$. The RH quantifier is $\forall f$, so U_f is unbounded and the honest object is $R^\dagger$ of the full cap (or a cofinal selected window large enough to cover f). The r450 two-chart split stands with the opposite moral: the prefix chart is vacuous; the full-window chart is load-bearing (kz17: $\delta_{\mathrm{pre}} = 0.21432$, $\delta_{\mathrm{full}} = 0.10864$).*

Against the finite tent $U_f = 2$, $n_f/N_w \in [0.247, 0.311]$ on the sealed r447/r448 windows, and every recorded flip sits past n_f (kz197: flip $3788 > n_f = 1175$). `TAIL_ONLY` (r449) survives for this f. It does not survive the $\forall f$ quantifier: that demand is the full window, on which kz197 is `EXACT_DEAD`.

M_∂ grows like $0.79 \log N$ because the *archimedean* window mass grows. It is equal in MAIN and ARCH. It is not von Mangoldt.

4 Q4 — Lean repair and retyping

Proposition 3 (minimal repair). *Retract `Iff.rfl` as a compression theorem. It is a naming of the full-cap cone. A prefix PSD statement is not the mincut and does not imply Weil positivity for any f with support past $\log 2$. The load-bearing named Prop remains `frequently_selected_augDualResolvent_ge_half` at W.`cap`.*

Retype, as a dictionary of the prime-blind zone (r303), not as mincut progress: r449 n_{stab}-location of ABD flips; r450 two-chart census plus the false identification; r451 $q^\dagger$-plateau; r452 $q_* = M_\partial/(M_\partial + 5/7)$; r453 $q_n = S_{n-1}/B_w$ and M_∂ growth; r454 $m_\infty = m_{\mathrm{ARCH}}$; r455 MAIN=ARCH for $j \leq J_P - 2$ (the SATZ that *makes* the prefix vacuous).

5 Kills

claim	verdict
prefix mincut $\Rightarrow$ RH path	**killed** (world-blind)
`Iff.rfl` $= n_{\mathrm{stab}}$-compression	**killed** (naming)
r455 eventually-per-n covers n_f	**killed** (fixed n)
M_∂ is von Mangoldt	**killed** (arch mass)
`TAIL_ONLY` vs $U_f = 2$	stands
`TAIL_ONLY` vs $\forall f$	fails (full window)
Lean full-cap cone is prime-blind	no (combHankel + `combRead`)

No RH claim. No anti-RH claim.

References

[1] L2 atom-target / reduction dictionary.

[2] Extraction red-team.

[3] Vacuous overflow.

[4] Flip versus stabilization.

[5] Prefix mincut.

[6] ARCH chain.